

① $|\vec{n}|=1$; pokaži $\vec{n} \times (\nabla \times \vec{n}) = -(\vec{n} \cdot \nabla) \vec{n}$ (z indeksi)

$$[\vec{n} \times (\nabla \times \vec{n})]_i = \epsilon_{ijk} n_j (\nabla \times \vec{n})_k = \epsilon_{ijk} n_j \epsilon_{klm} (\partial_l n_m) = \epsilon_{kij} n_j \epsilon_{klm} (\partial_l n_m) =$$

$$(\vec{a} \times \vec{b})_i = \epsilon_{ijk} a_j b_k$$

predznak permutacije

$$\boxed{\epsilon_{kij} \epsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}}$$

$$\delta_{ij} n_j = n_i$$

$$= (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) n_j (\partial_l n_m) = \underbrace{n_j (\partial_l n_j)}_0 - n_j (\partial_j n_i) = \dots \checkmark$$

Kodre 3.11

$$U = \frac{[(\vec{p} \times \vec{r}) \times \vec{r}] \cdot \vec{p}}{r^{m+2}}, \quad V = \frac{p^2}{r^m}$$

za katere m, k bo za $\varphi = U + kV$ $\nabla^2 \varphi = 0$?

$$\partial_i V = p^2 \partial_i r^{-m} = -m p^2 r_i r^{-m-2}$$

$$\partial_i \partial_i V = -m p^2 (3 r^{-m-2} - (m+2) r_i^2 r^{-m-4}) = \frac{m(m-1) p^2 r^{-m-2}}{}$$

$$\hookrightarrow \partial_i r_i = \partial_x r_x + \partial_y r_y + \partial_z r_z = 1+1+1=3$$

$$\partial_i U = \partial_i \frac{\epsilon_{jkl} p_k r_l \epsilon_{ajb} r_b p_a}{r^{m+2}} = \epsilon_{jkl} \epsilon_{ajb} p_k p_a \partial_i \frac{r_l r_b}{r^{m+2}}$$

$$\partial_i \frac{r_l r_b}{r^{m+2}} = (\partial_i r_l) \frac{r_b}{r^{m+2}} + (\partial_i r_b) \frac{r_l}{r^{m+2}} + r_b r_l \partial_i r^{-m-2} = \delta_{il} \frac{r_b}{r^{m+2}} + \delta_{ib} \frac{r_l}{r^{m+2}} + r_b r_l (-m-2) r_i r^{-m-4}$$

$$\partial_i \partial_i \frac{r_l r_b}{r^{m+2}} = \delta_{il} \delta_{ib} r^{-m-2} + \delta_{il} r_b r_i (-m-2) r^{-m-4} + \delta_{ib} \delta_{il} r^{-m-2} + \delta_{ib} r_l r_i (-m-2) r^{-m-4} -$$

$$-(m+2) [\delta_{ib} r_l r_i r^{-m-4} + \delta_{il} r_b r_i r^{-m-4} + 3 r_l r_b r_i r^{-m-4} + (-m-4) r_i r_l r_b r_i r^{-m-6}] =$$

$$= r^{-m-2} [\delta_{il} \delta_{ib} + \delta_{ib} \delta_{il}] - (m+2) r^{-m-4} [\underbrace{\delta_{il} r_b r_i}_{r_l r_b} + \underbrace{\delta_{ib} r_l r_i}_{r_l r_b} + \underbrace{\delta_{ib} r_l r_i}_{r_l r_b} + \underbrace{\delta_{il} r_l r_b}_{r_l r_b} + 3 r_l r_b - (m+4) r_l r_b] = r^{-m-2} [2 \delta_{ib}] - (m+2) r^{-m-4} [(7-m-4) r_l r_b]$$

$$\partial_i \partial_i U = \xi_{jkl} \xi_{ajb} p_k p_a \partial_i \partial_i \frac{r_i r_b}{r^{m+2}} = \xi_{jkl} \xi_{jba} p_k p_a \underline{\hspace{1cm}}$$

$$= (\delta_{kb} \delta_{la} - \delta_{ka} \delta_{lb}) p_k p_a [r^{m-2} 2 \delta_{lb} + (m+2)(m-3) r^{m-4} r_i r_b] =$$

$$= (p_b p_l - \delta_{lb} p_k p_k) (r^{m-2} 2 \delta_{lb} + (m+2)(m-3) r^{m-4} r_i r_b) =$$

$$= \underline{r^{m-2} \cdot 2 p^2} + (m+2)(m-3) r^{m-4} p_b r_b p_l r_l - \underline{p^2 r^{m-2} \cdot 2 \cdot 3} \overset{\delta_{lb} \delta_{lb}}{\underline{-(m+2)(m-3) p^2 r^{m-2}}} =$$

$$= -((m+2)(m-3) + 4) \frac{p^2}{r^{m+2}} + (m+2)(m-3) r^{m-4} \underbrace{p_b p_l r_b r_l}_{(\vec{p} \cdot \vec{r})^2}$$

$$\nabla^2 \varphi = \nabla^2 U + k \nabla^2 V = 0$$

$$\underline{-((m+2)(m-3) + 4) \frac{p^2}{r^{m+2}}} + (m+2)(m-3) \underline{\frac{(\vec{p} \cdot \vec{r})^2}{r^{m+4}}} + k \underline{\frac{m(m-1)p^2}{r^{m+2}}} = 0$$

$$\therefore k m(m-1) = \cancel{(m+2)(m-3)} + 4$$

$$\therefore (m+2)(m-3) = 0$$

$$k = \frac{4}{m(m-1)} = \begin{cases} \frac{2}{3}; m = -2 \\ \frac{2}{3}; m = 3 \end{cases}$$

$$\begin{aligned} m_1 &= -2 \\ m_2 &= 3 \end{aligned}$$

$$\varphi(k = \frac{2}{3}, m = -2) = (\vec{r} \cdot \vec{p})^2 - r^2 p^2 + \frac{2}{3} p^2 r^2 = (\vec{r} \cdot \vec{p})^2 - \frac{1}{3} p^2 r^2$$

• sedlast potencial

$$\varphi(k = \frac{2}{3}, m = 3) = \frac{(\vec{r} \cdot \vec{p})^2 - r^2 p^2}{r^5} + \frac{2}{3} \frac{p^2}{r^3} = \frac{(\vec{r} \cdot \vec{p})^2 - \frac{1}{3} p^2 r^2}{r^5}$$

56) Hitrostno polje okoli dolgega cilindra za $Re \gg 1$



$$\vec{v} = \nabla U, \quad U = v_0 x \left(1 + \frac{1}{2} \left(\frac{v_0}{r} \right)^3 \right)$$

a) pokaži, da velja $v_r|_{r=r_0} = 0$

b) izračunaj $p(\varphi)|_{r=r_0} = ?$

$$\begin{aligned} v_r &= (\hat{e}_r \cdot \nabla) U = \frac{\partial U}{\partial r} = \frac{\partial}{\partial r} \left(v_0 r \cos \varphi \left(1 + \frac{1}{2} \left(\frac{v_0}{r} \right)^3 \right) \right) = v_0 \cos \varphi + \frac{1}{2} v_0^3 (-2 r^{-3}) \cos \varphi = \\ &= v_0 \cos \varphi \left(1 - \left(\frac{v_0}{r} \right)^3 \right) \end{aligned}$$

$$v_r|_{r=r_0} = 0 \quad \checkmark$$